

# Introduction

In the modern world of electronics, the term **Digital** is generally associated with a computer because the term **Digital** is derived from the way computers perform operation, by counting digits. For many years, the application of digital electronics was only in the computer system. But now-a-days, digital electronics is used in many other applications. Following are some of the examples in which **Digital electronics** is heavily used.

- Industrial process control
- Military system
- Television
- Communication system
- Medical equipment
- Radar
- Navigation

## Signal

**Signal** can be defined as a physical quantity, which contains some information. It is a function of one or more than one independent variables. Signals are of two types.

- Analog Signal
- Digital Signal

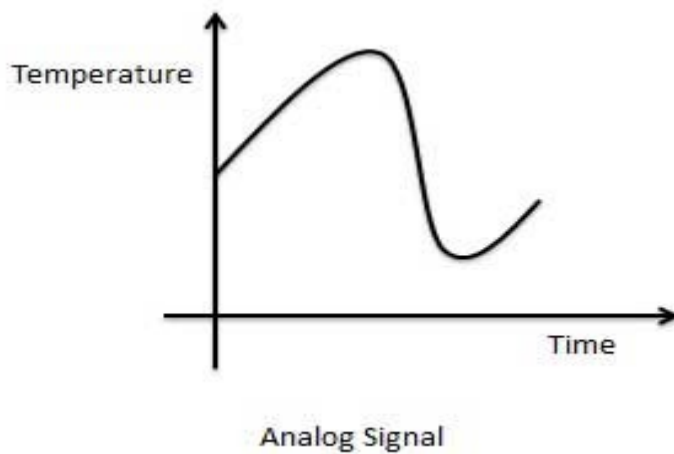
## Analog Signal

An **analog signal** is defined as the signal having continuous values. Analog signal can have infinite number of different values. In real world scenario, most of the things observed in nature are analog. Examples of the analog signals are following.

- Temperature
- Pressure

- Distance
- Sound
- Voltage
- Current
- Power

### Graphical representation of Analog Signal (Temperature)



The circuits that process the analog signals are called as analog circuits or system. Examples of the analog system are following.

- Filter
- Amplifiers
- Television receiver
- Motor speed controller

### Disadvantage of Analog Systems

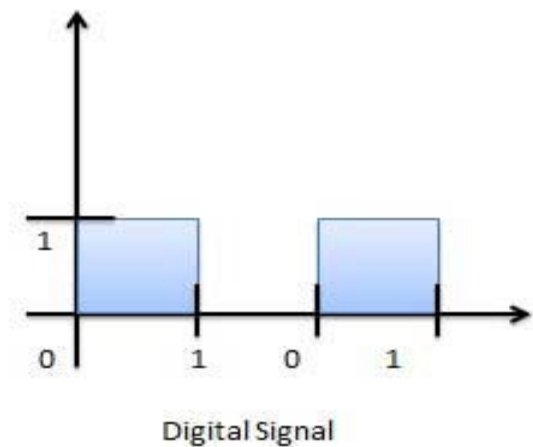
- Less accuracy
- Less versatility
- More noise effect
- More distortion

# Digital Signal

A **digital signal** is defined as the signal which has only a finite number of distinct values. Digital signals are not continuous signals. In the digital electronic calculator, the input is given with the help of switches. This input is converted into electrical signal which have two discrete values or levels. One of these may be called low level and another is called high level. The signal will always be one of the two levels. This type of signal is called digital signal. Examples of the digital signal are following.

- Binary Signal
- Octal Signal
- Hexadecimal Signal

## Graphical representation of the Digital Signal (Binary)



The circuits that process the digital signals are called digital systems or digital circuits. Examples of the digital systems are following.

- Registers
- Flip-flop
- Counters

- Microprocessors

## Advantage of Digital Systems

- More accuracy
- More versatility
- Less distortion
- Easy communicate
- Possible storage of information

## Comparison of Analog and Digital Signal

| S.N. | Analog Signal                                                                                                                                                                                                                                                             | Digital Signal                                   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| 1    | Analog signal has infinite values.                                                                                                                                                                                                                                        | Digital signal has a finite number of values.    |
| 2    | Analog signal has a continuous nature.                                                                                                                                                                                                                                    | Digital signal has a discrete nature.            |
| 3    | Analog signal is generated by transducers(A <b>transducer</b> is an electronic device that converts energy from one form to another. Common examples include microphones, loudspeakers, thermometers, position and pressure sensors, and antenna.) and signal generators. | Digital signal is generated by A to D converter. |
| 4    | Example of analog signal – sine wave, triangular waves.                                                                                                                                                                                                                   | Example of digital signal – binary signal.       |

## Digital Number System

A digital system can understand positional number system only where there are a few symbols called digits and these symbols represent different values depending on the position they occupy in the number.

A value of each digit in a number can be determined using

- The digit
- The position of the digit in the number
- The base of the number system (where base is defined as the total number of digits available in the number system).

## Decimal Number System

The number system that we use in our day-to-day life is the decimal number system. Decimal number system has base 10 as it uses 10 digits from 0 to 9. In decimal number system, the successive positions to the left of the decimal point represent units, tens, hundreds, thousands and so on.

Each position represents a specific power of the base (10). For example, the decimal number 1234 consists of the digit 4 in the units position, 3 in the tens position, 2 in the hundreds position, and 1 in the thousands position, and its value can be written as

$$\begin{aligned} &(1 \times 1000) + (2 \times 100) + (3 \times 10) + (4 \times 1) \\ &(1 \times 10^3) + (2 \times 10^2) + (3 \times 10^1) + (4 \times 10^0) \\ &1000 + 200 + 30 + 4 \\ &1234 \end{aligned}$$

As a computer programmer or an IT professional, you should understand the following number systems which are frequently used in computers.

| S.N. | Number System & Description |
|------|-----------------------------|
|------|-----------------------------|

|   |                                                                                           |
|---|-------------------------------------------------------------------------------------------|
|   |                                                                                           |
| 1 | <b>Binary Number System</b><br><br>Base 2. Digits used: 0, 1                              |
| 2 | <b>Octal Number System</b><br><br>Base 8. Digits used: 0 to 7                             |
| 3 | <b>Hexa Decimal Number System</b><br><br>Base 16. Digits used: 0 to 9, Letters used: A- F |

## Binary Number System

### Characteristics

- Uses two digits, 0 and 1.
- Also called base 2 number system
- Each position in a binary number represents a 0 power of the base (2). Example:  $2^0$

### Example

Binary Number:  $10101_2$

Calculating Decimal Equivalent –

| Step   | Binary Number | Decimal Number                                                                              |
|--------|---------------|---------------------------------------------------------------------------------------------|
| Step 1 | $10101_2$     | $((1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0))_{10}$ |
| Step 2 | $10101_2$     | $(16 + 0 + 4 + 0 + 1)_{10}$                                                                 |
| Step 3 | $10101_2$     | $21_{10}$                                                                                   |

|  |  |  |
|--|--|--|
|  |  |  |
|--|--|--|

**Note:** 10101<sub>2</sub> is normally written as 10101.

## Octal Number System

### Characteristics

- Uses eight digits, 0,1,2,3,4,5,6,7.
- Also called base 8 number system
- Each position in an octal number represents a 0 power of the base (8). Example: 8<sup>0</sup>

### Example

Octal Number – 12570<sub>8</sub>

Calculating Decimal Equivalent –

| Step   | Octal Number       | Decimal Number                                                                              |
|--------|--------------------|---------------------------------------------------------------------------------------------|
| Step 1 | 12570 <sub>8</sub> | $((1 \times 8^4) + (2 \times 8^3) + (5 \times 8^2) + (7 \times 8^1) + (0 \times 8^0))_{10}$ |
| Step 2 | 12570 <sub>8</sub> | $(4096 + 1024 + 320 + 56 + 0)_{10}$                                                         |
| Step 3 | 12570 <sub>8</sub> | 5496 <sub>10</sub>                                                                          |

**Note:** 12570<sub>8</sub> is normally written as 12570.

# Hexadecimal Number System

## Characteristics

- Uses 10 digits and 6 letters, 0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F.
- Letters represents numbers starting from 10. A = 10, B = 11, C = 12, D = 13, E = 14, F = 15.
- Also called base 16 number system.
- Each position in a hexadecimal number represents a 0 power of the base (16).  
Example  $16^0$ .

## Example –

Hexadecimal Number:  $19FDE_{16}$

## Calculating Decimal Equivalent –

| Step   | Hexadecimal Number | Decimal Number                                                                                      |
|--------|--------------------|-----------------------------------------------------------------------------------------------------|
| Step 1 | $19FDE_{16}$       | $((1 \times 16^4) + (9 \times 16^3) + (F \times 16^2) + (D \times 16^1) + (E \times 16^0))_{10}$    |
| Step 2 | $19FDE_{16}$       | $((1 \times 16^4) + (9 \times 16^3) + (15 \times 16^2) + (13 \times 16^1) + (14 \times 16^0))_{10}$ |
| Step 3 | $19FDE_{16}$       | $(65536 + 36864 + 3840 + 208 + 14)_{10}$                                                            |
| Step 4 | $19FDE_{16}$       | $106462_{10}$                                                                                       |

**Note** –  $19FDE_{16}$  is normally written as 19FDE.



## Lecture 2:

### Number System Conversion

There are many methods or techniques which can be used to convert numbers from one base to another. We'll demonstrate here the following –

- Decimal to Other Base System
- Other Base System to Decimal
- Other Base System to Non-Decimal
- Shortcuts method – Binary to Octal
- Shortcuts method – Octal to Binary
- Shortcuts method – Binary to Hexadecimal
- Shortcuts method – Hexadecimal to Binary

### Decimal to Other Base System

Steps

- **Step 1** – Divide the decimal number to be converted by the value of the new base.
- **Step 2** – Get the remainder from Step 1 as the rightmost digit (least significant digit) of new base number.
- **Step 3** – Divide the quotient of the previous divide by the new base.
- **Step 4** – Record the remainder from Step 3 as the next digit (to the left) of the new base number.

Repeat Steps 3 and 4, getting remainders from right to left, until the quotient becomes zero in Step 3.

The last remainder thus obtained will be the Most Significant Digit (MSD) of the new base number.

Example –

Decimal Number:  $29_{10}$

### Calculating Binary Equivalent –

| Step   | Operation | Result | Remainder |
|--------|-----------|--------|-----------|
| Step 1 | 29 / 2    | 14     | 1         |
| Step 2 | 14 / 2    | 7      | 0         |
| Step 3 | 7 / 2     | 3      | 1         |
| Step 4 | 3 / 2     | 1      | 1         |
| Step 5 | 1 / 2     | 0      | 1         |

As mentioned in Steps 2 and 4, the remainders have to be arranged in the reverse order so that the first remainder becomes the Least Significant Digit (LSD) and the last remainder becomes the Most Significant Digit (MSD).

Decimal Number –  $29_{10}$  = Binary Number –  $11101_2$ .

### Other Base System to Decimal System

#### Steps

- **Step 1** – Determine the column (positional) value of each digit (this depends on the position of the digit and the base of the number system).
- **Step 2** – Multiply the obtained column values (in Step 1) by the digits in the corresponding columns.
- **Step 3** – Sum the products calculated in Step 2. The total is the equivalent value in decimal.

### Example

Binary Number –  $11101_2$

Calculating Decimal Equivalent –

| Step   | Binary Number | Decimal Number                                                                              |
|--------|---------------|---------------------------------------------------------------------------------------------|
| Step 1 | $11101_2$     | $((1 \times 2^4) + (1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0))_{10}$ |
| Step 2 | $11101_2$     | $(16 + 8 + 4 + 0 + 1)_{10}$                                                                 |
| Step 3 | $11101_2$     | $29_{10}$                                                                                   |

Binary Number –  $11101_2$  = Decimal Number –  $29_{10}$

## Other Base System to Non-Decimal System

Steps

- **Step 1** – Convert the original number to a decimal number (base 10).
- **Step 2** – Convert the decimal number so obtained to the new base number.

## Example

Octal Number –  $25_8$

Calculating Binary Equivalent –

### Step 1 – Convert to Decimal

| Step   | Octal Number | Decimal Number                           |
|--------|--------------|------------------------------------------|
| Step 1 | $25_8$       | $((2 \times 8^1) + (5 \times 8^0))_{10}$ |
| Step 2 | $25_8$       | $(16 + 5)_{10}$                          |

|        |        |           |
|--------|--------|-----------|
| Step 3 | $25_8$ | $21_{10}$ |
|--------|--------|-----------|

Octal Number –  $25_8$  = Decimal Number –  $21_{10}$

## Step 2 – Convert Decimal to Binary

| Step   | Operation | Result | Remainder |
|--------|-----------|--------|-----------|
| Step 1 | $21 / 2$  | 10     | 1         |
| Step 2 | $10 / 2$  | 5      | 0         |
| Step 3 | $5 / 2$   | 2      | 1         |
| Step 4 | $2 / 2$   | 1      | 0         |
| Step 5 | $1 / 2$   | 0      | 1         |

Decimal Number –  $21_{10}$  = Binary Number –  $10101_2$

Octal Number –  $25_8$  = Binary Number –  $10101_2$

## Shortcuts method - Binary to Octal

Steps

- **Step 1** – Divide the binary digits into groups of three (starting from the right).
- **Step 2** – Convert each group of three binary digits to one octal digit.

## Example

Binary Number –  $10101_2$

Calculating Octal Equivalent –

| Step   | Binary Number      | Octal Number                  |
|--------|--------------------|-------------------------------|
| Step 1 | 10101 <sub>2</sub> | 010 101                       |
| Step 2 | 10101 <sub>2</sub> | 2 <sub>8</sub> 5 <sub>8</sub> |
| Step 3 | 10101 <sub>2</sub> | 25 <sub>8</sub>               |

Binary Number – 10101<sub>2</sub> = Octal Number – 25<sub>8</sub>

## Shortcuts method - Octal to Binary

Steps

- **Step 1** – Convert each octal digit to a 3 digit binary number (the octal digits may be treated as decimal for this conversion).
- **Step 2** – Combine all the resulting binary groups (of 3 digits each) into a single binary number.

## Example

Octal Number – 25<sub>8</sub>

Calculating Binary Equivalent –

| Step   | Octal Number    | Binary Number                     |
|--------|-----------------|-----------------------------------|
| Step 1 | 25 <sub>8</sub> | 2 <sub>10</sub> 5 <sub>10</sub>   |
| Step 2 | 25 <sub>8</sub> | 010 <sub>2</sub> 101 <sub>2</sub> |
| Step 3 | 25 <sub>8</sub> | 010101 <sub>2</sub>               |

Octal Number –  $25_8$  = Binary Number –  $10101_2$

## Shortcut method - Binary to Hexadecimal

Steps

- **Step 1** – Divide the binary digits into groups of four (starting from the right).
- **Step 2** – Convert each group of four binary digits to one hexadecimal symbol.

## Example

Binary Number –  $10101_2$

Calculating hexadecimal Equivalent –

| Step   | Binary Number | Hexadecimal Number |
|--------|---------------|--------------------|
| Step 1 | $10101_2$     | 0001 0101          |
| Step 2 | $10101_2$     | $1_{10}$ $5_{10}$  |
| Step 3 | $10101_2$     | $15_{16}$          |

Binary Number –  $10101_2$  = Hexadecimal Number –  $15_{16}$

## Shortcut method - Hexadecimal to Binary

Steps

- **Step 1** – Convert each hexadecimal digit to a 4 digit binary number (the hexadecimal digits may be treated as decimal for this conversion).

- **Step 2** – Combine all the resulting binary groups (of 4 digits each) into a single binary number.

## Example

Hexadecimal Number –  $15_{16}$

Calculating Binary Equivalent –

| Step   | Hexadecimal Number | Binary Number   |
|--------|--------------------|-----------------|
| Step 1 | $15_{16}$          | $1_{10} 5_{10}$ |
| Step 2 | $15_{16}$          | $0001_2 0101_2$ |
| Step 3 | $15_{16}$          | $00010101_2$    |

Hexadecimal Number –  $15_{16}$  = Binary Number –  $10101_2$